

AI-Enhanced X-ray Computed Tomography for Microelectronics: A Brief Review of Machine Learning and Deep Learning Integration in Non-Destructive Failure Analysis

Marina Adriana Mercioni
Applied Electronics Department
Politehnica University Timisoara
Timisoara, Romania
marina.mercioni@upt.ro

Cătălin Daniel Căleanu
Applied Electronics Department
Politehnica University Timisoara
Timisoara, Romania
catalin.caleanu@upt.ro

Raul Ciprian Ionel
Measurements and Optical Electronics
Department
Politehnica University Timisoara
Timisoara, Romania
raul.ionel@upt.ro

Abstract—This comprehensive review investigates the intersection of Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL) in X-ray Computed Tomography (XCT), focusing on non-destructive testing (NDT) and microelectronics failure analysis. As electronic components continue to shrink, conventional inspection methods become increasingly insufficient due to their inability to accurately detect subtle defects. Recent advancements in AI architectures, such as 3D Convolutional Neural Networks (3D CNNs), YOLO variants, and Vision Transformers, have significantly improved the detection of microscopic defects like voids and cracks, while also enhancing three-dimensional image reconstruction capabilities. These data-driven innovations lead to substantial reductions in acquisition times and mitigate inherent imaging artifacts including noise and beam hardening effects. However, this review examines current industrial deployment challenges, primarily centered on the scarcity of annotated datasets for training models, poor model generalizability across diverse hardware configurations, high computational costs associated with DL algorithms, and the lack of transparency surrounding their decision-making processes. To bridge the gap between academic research and practical applications in industry, we identify future directions that include Explainable AI (XAI) techniques to enhance decision transparency, synthetic data generation methods for improved model training, and Parameter-Efficient Fine-Tuning (PEFT).

Keywords— *X-ray computed tomography, microelectronics, artificial intelligence, machine learning, deep learning, non-destructive testing.*

I. INTRODUCTION

The microelectronics industry's rapid evolution has led to increased interconnect density and reduced critical dimensions in component miniaturization and complex packaging architectures [1][2][3]. This complexity poses significant

challenges for ensuring electronic device quality and reliability, with hidden defects in scaled-down interconnects being a primary source of production yield reduction.

Traditional failure analysis methods are time-consuming, labor-intensive, prone to errors, and limited to 2D visualization. Boundary Scan Test (IEEE Std. 1149.1) is one feasible non-invasive solution for addressing situations where physical access to electronic board nets is restricted [4]. However, its limitations necessitate complementary technologies. To overcome these challenges, XCT and 3D X-ray Microscopy have become essential tools for non-destructive internal structure and defect visualization in microelectronics [5][6][7]. Despite the benefits of XCT technology, it faces challenges such as slow high-resolution volumetric scans, image quality issues due to noise, ray scattering, and beam hardening artifacts [1][3][8].

The integration of AI, ML, and DL methods promises to revolutionize radiological inspection by automating analysis and streamlining workflows [3]. DL algorithms transform the failure analysis workflow through automatic feature extraction from images for multiscale defect detection and component identification, as well as improved 3D reconstruction quality. This accelerates scanning processes and reduces noise in data acquisitions with limited projections or low doses [1][7][9].

This review aims to provide a comprehensive evaluation of AI-based applications in non-destructive failure analysis using XCT. It highlights the importance of these intelligent technologies in addressing practical challenges, exploring state-of-the-art solutions for XCT image reconstruction, automated defect detection, and advanced component segmentation [10]. The review identifies current limitations and outlines future research directions.

The paper is organized into five sections. Section 2 presents the research methodology, while Section 3 discusses related work. Section 4 covers discussions of the findings, and finally, in Section 5, we conclude with a summary of key points.

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II. RESEARCH METHODOLOGY

This review aims to provide an up-to-date assessment of AI approaches in computed tomography (CT) image reconstruction, segmentation, and analysis with XCT, as well as other NDT techniques. The investigation seeks to address the following research questions (RQ):

- RQ1: What are the most advanced AI methods, models, and algorithms currently employed for image reconstruction and defect detection?
- RQ2: What are the primary practical applications and best use cases of these technologies in industry?
- RQ3: Which public or private datasets are available for training, testing, and validating DL models in this domain?
- RQ4: What are the major challenges, limitations, and problems that limit the potential of AI applications?
- RQ5: Which future directions, trends, and technological innovations can be expected in applying AI to industrial CT?

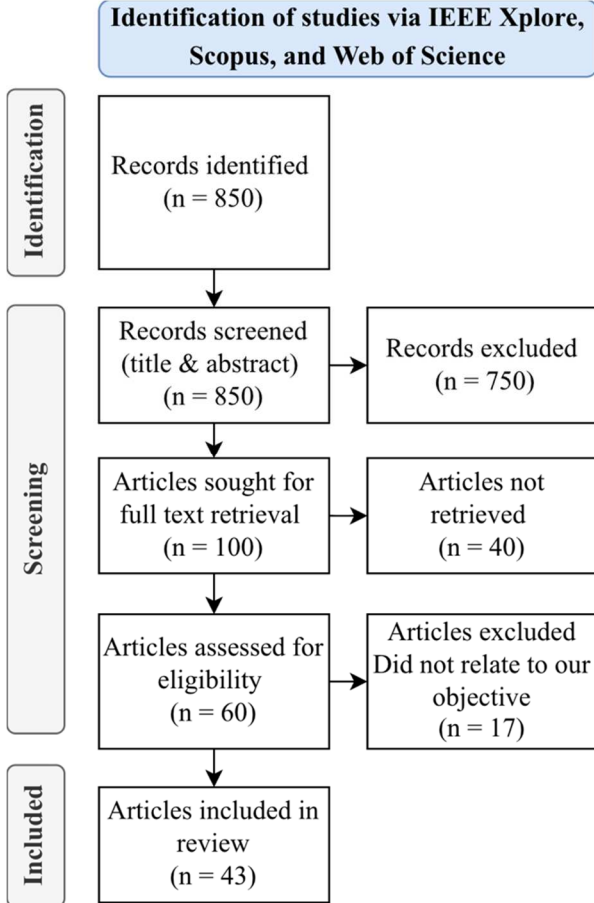


Fig. 1. Flow diagram of the systematic search and selection process conducted via IEEE Xplore, Scopus, and Web of Science for studies on AI-enhanced XCT for microelectronics in the automotive industry.

A comprehensive search was conducted across three electronic databases (IEEE Xplore, Scopus, and Web of

Science), focusing on articles published between 2010 and 2026. Search terms included “X-ray computed tomography”, “microelectronics”, “artificial intelligence”, “machine learning”, “deep learning”, “non-destructive testing”, and specifically, the application of “deep learning for noise reduction in imaging”. Boolean operators (AND/OR) were employed to refine search queries and ensure that relevant studies were retrieved (see Fig. 1).

To ensure data quality and relevance, a set of inclusion criteria was established: studies had to be peer-reviewed journal articles or conference papers published in English from 2010 onwards with explicit applications of AI methodologies in domains such as CT image reconstruction, defect detection, or automotive manufacturing. Studies featuring quantitative metrics on model accuracy, precision, effectiveness, and practical use cases were selected.

Additionally, some studies required clinical validation or comparison with conventional baselines to evaluate parameters like image quality and radiation dose [11][12]. Exclusion criteria applied universally included non-English publications, non-peer-reviewed materials, editorials, case reports, conference abstracts, traditional image-processing methods without AI applications, limited full-text availability, and insufficient data for robust quantitative analysis.

III. RELATED WORK

This chapter addresses research questions RQ1-RQ5 by providing an in-depth examination of cutting-edge methods and techniques for XCT data analysis and their practical applications.

A. State-of-the-Art Methods and Algorithms (RQ1)

The literature reveals a massive shift from traditional image processing toward ML and DL for both image reconstruction and defect detection in XCT [13]. DL Reconstruction (DLR) and DL Iterative Reconstruction (DLIR) are widely employed to improve XCT image quality while reducing scan times and radiation doses [14]. Algorithms utilizing CNNs and Generative Adversarial Networks (GANs) effectively reduce image noise and correct beam-hardening artifacts, enabling up to a 10x improvement in throughput for high-resolution scans [7]. The U-Net architecture (and its 2D/3D variants) is the predominant choice for semantic and instance segmentation, particularly for delineating intricate boundaries in composite materials, Printed Circuit Boards (PCBs), and medical imaging [2][15][16][17][18][19][20]. For real-time object and defect detection, You Only Look Once (YOLO) variants (e.g., YOLOv3, YOLOv4, YOLOv5, and YOLOSeg) are heavily favored due to their optimal balance between speed and precision [16][21][22].

Additionally, advanced architectures such as Mask R-CNN, SegFormer, and Vision Transformers (ViTs) and Segment Anything Models (SAM/TGSAM) are increasingly adopted to extract global context and improve multi-scale segmentation accuracy [2][23][24][25][26]. A new paradigm is the use of ensemble and multi-method approaches, combining multiple AI models (e.g., CNN-LSTM [Long Short-Term Memory]) to overcome the trade-offs of single-method deployments [23][27].

B. Practical Applications and Use Cases (RQ2)

AI-enhanced XCT and NDT have demonstrated high accuracy and robustness across various industrial sectors such as microelectronics and PCB inspection, automotive manufacturing, aerospace and composite materials, civil infrastructure. AI models successfully automate the detection of hidden defects in advanced chip packaging, such as voids in solder joints, Through Silicon Vias (TSVs), and Ball Grid Arrays (BGAs) [1][3][28][29]. DL frameworks are also applied to reverse engineering and Hardware Assurance (HA), allowing for automated 3D semantic segmentation of PCB layers, netlist extraction, and counterfeit detection [6][17][30] (see Fig. 2). AI applications focus heavily on production optimization, quality assurance, and assembly monitoring. Use cases include detecting surface defects on engine blocks, predicting screw-fastening quality, assessing weld quality, and implementing predictive maintenance to forecast machine breakdowns [27][31]. For materials like Carbon Fiber-Reinforced Polymers (CFRP) and Glass Fiber-Reinforced Polymers (GFRP), DL algorithms applied to Micro-CT data can automatically segment and quantify internal pore distributions, delamination, and matrix cracks, which are critical for structural integrity [32][33][34][35]. AI-enhanced Ground Penetrating Radar (GPR) and acoustic-laser techniques are utilized to detect structural defects like concrete cracks, subsurface voids in airport runways, and hidden corrosion [36].

C. Available Datasets for Training and Validation (RQ3)

The availability of high-quality datasets is a critical factor in model performance, and their impact varies across different domains. In DL-based defect detection and XCT reconstruction, researchers utilize a combination of publicly available benchmarks and their own industrial datasets for training and validation. Notable public resources include the FICS PCB X-ray dataset, designed specifically for automated PCB inspection, as well as the NIST XCT dataset, which provides high-resolution scans of additively manufactured (AM) cobalt chrome samples [3][6] (Table I). Moreover, medical and general computer vision datasets such as NIH-AAPM-Mayo clinical CT (2016), LUNA16, LIDC-IDRI, ImageNet, and COCO are extensively employed [14]. Despite this, the limited availability of open-source data persists due to data privacy concerns and the

high cost of expert annotation, leading many studies to rely heavily on private, custom-acquired datasets. Examples of such private datasets include the CIDX-Ray1950 dataset for detecting chip packaging void defects [2], HBM X-ray microscopy scans, and undisclosed industrial datasets containing wafer die defects or short glass fiber reinforced polymer composite scans [21][23].

D. Major Problems, Limitations, and Challenges (RQ4)

Despite recent breakthroughs, several barriers limit the full potential of AI such as acquiring large volumes of anomalous/defective industrial samples is highly resource-intensive [3]; model generalization issues, as algorithms trained on specific experimental setups frequently encounter low accuracy when applied to real-world industrial environments due to variations in lighting, material density, object orientation, or scanning speeds [16]; and high computational costs associated with advanced 3D CNNs and Transformer models that demand massive computational power, memory, and training time [1][27]. The “blackbox” due to the absence of interpretability in DL models creates a barrier to trust and regulatory compliance in critical fields like automotive, aerospace and medicine [37].

E. Future Directions and Innovations (RQ5)

Several future directions include XAI approaches that make AI decisions transparent using following feature attribution methods such as SHapley Additive exPlanations (SHAP), Gradient-weighted Class Activation Mapping (GradCAM), and Local Interpretable Model-agnostic Explanations (LIME) [23][37]; semi-supervised and unsupervised learning techniques like active learning, weak supervision, and autoencoders that learn to detect anomalies purely from normal data [16][38][39]; parameter-efficient fine-tuning methods such as Low-Rank Adaptation (LoRA) that are being applied to large foundation models (e.g., SAM) to adapt them for specific industrial tasks, such as PCB component segmentation, using very few annotated examples (few-shot learning) [24]; and multi-modal and hardware integration approaches that fuse XCT data with other NDT modalities (e.g., infrared thermography or acoustic emission, Computed Laminography), leveraging edge-computing and quantum computing to achieve real-time, closed-loop decision making [23][26][32][40].

TABLE I: COMPARATIVE SUMMARY OF AI TECHNIQUES IN XCT DEFECT ANALYSIS

Ref	Research Focus	Method/Model	Performance
[15], [29]	Defect detection and semantic segmentation in XCT slices AM components and PCB solder joints.	U-Net and 3D CNN	Robust capability in segmenting fine defects from metal matrix backgrounds; 3D CNNs achieved a high Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.9005 for volumetric defect classification.
[15], [21] 5/7/2026 6:25:00 PM	Real-time defect localization in X-ray imaging and wafer die particle instance segmentation.	YOLO Variants (e.g., YOLOv5s, YOLOSeg)	YOLOSeg achieved an Average Precision (AP) of 0.821 and an Intersection over Union (IoU) of 0.732 for highly miniaturized defects, while YOLOv5s reached a recall of 87.65% in AM defect detection.
[24]	Few-shot component segmentation in highly complex and densely packed PCB X-ray scans.	ViTs integrated within the SAM	PEFT with spatial adapters, the ViT-based model achieved an Average Dice Score of 0.711, significantly improving feature extraction over baseline zero-shot predictions.
[21]	Synthetic image augmentation to address severe data scarcity and class imbalance in semiconductor defect inspection.	GANs and Denoising Diffusion Probabilistic Models (DDPMs)	Integrating DDPM augmentation successfully diversified defect appearances, elevating the segmentation IoU from 0.715 to 0.754 and effectively mitigating overfitting risks despite the added computational overhead.

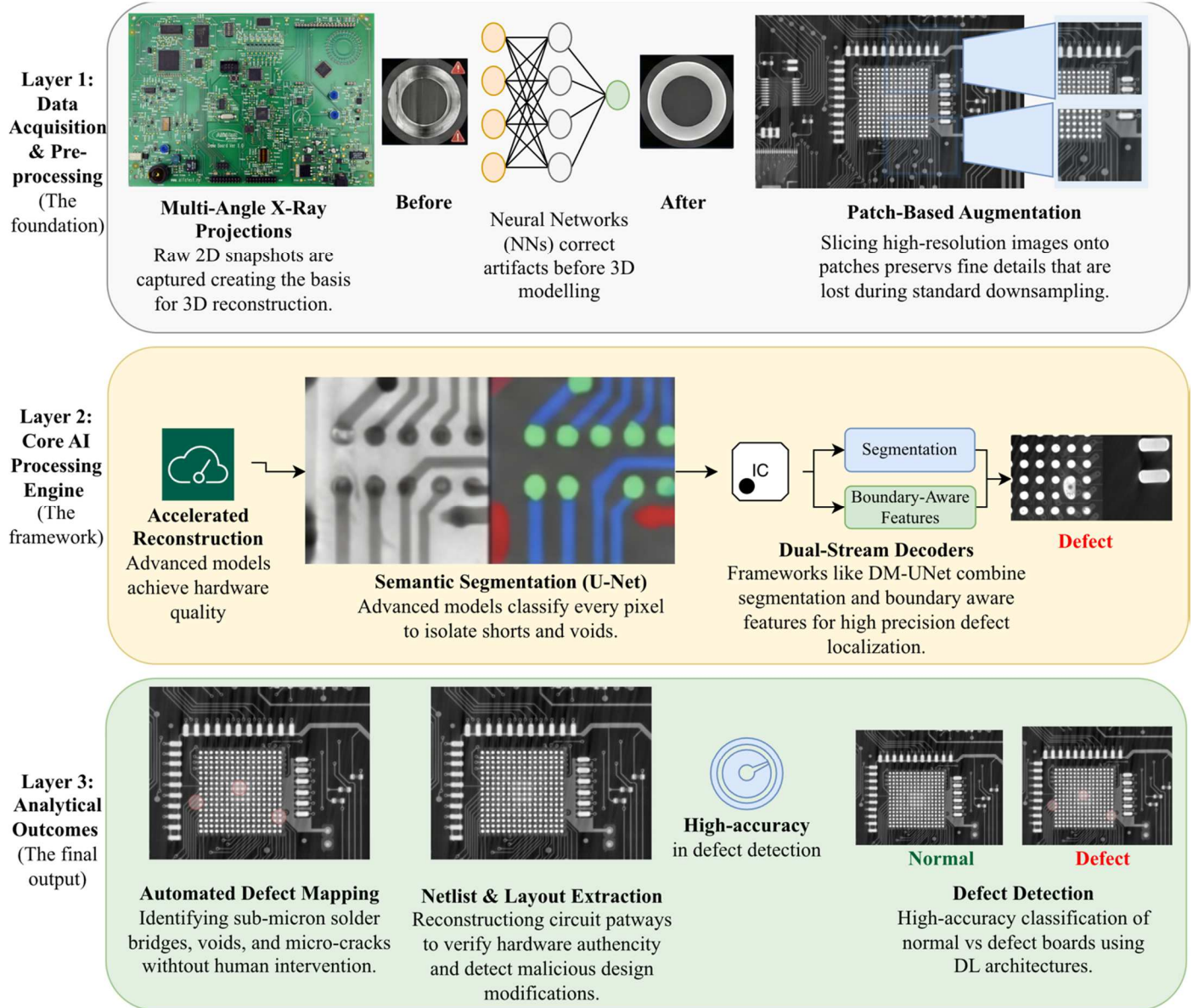


Fig. 2. The AI-driven XCT inspection pipeline features a multi-layered architecture for hardware assurance. 3D U-Net architecture: Layer 1 – Data Acquisition & Preprocessing; Layer 2 – Core AI Processing Engine; Layer 3 – Analytical Outcomes - A visualization of an XCT scan of a PCB and its defects, supported by Waygate Technologies, utilizing the V|tome|x S system.

IV. DISCUSSIONS

The development and application of AI, particularly for XCT failure analysis, dataset scale and diversity are pivotal factors in model training [16]. The findings underscore the necessity to ensure successful generalization across diverse real-world applications, necessitating datasets that not only comprise large quantities of samples but also exhibit representativeness and variability.

A. Critical Comparative Analysis of Existing AI Methodologies

A critical review of existing AI methodologies reveals a trade-off between segmentation precision, computational efficiency, and data requirements across different architectures

[16]. CNNs, such as U-Net and its 3D variants, excel at preserving fine spatial details but struggle to capture long-range global contexts and are computationally heavy for large volumetric scans [23].

In contrast, single-stage object detection models like YOLO offer real-time processing speeds but often exhibit lower recall rates and difficulty with small, densely packed micro-defects [14][21].

To address these limitations, instance segmentation models such as Mask R-CNN provide superior boundary delineation but have significant computational overhead, making them less viable for real-time deployment [23]. ViTs have been introduced to overcome CNN limitations by leveraging self-attention

mechanisms, offering superior accuracy despite their massive training datasets and quadratic computational costs [2][16]. To mitigate data scarcity, GANs and DDPM are increasingly used for synthetic data augmentation [21][41]. Simulators generating data from CAD models are also used to train networks where real-world defective samples are rare [38].

However, these generative models face challenges such as training instability, mode collapse, and the risk of introducing unrealistic image artifacts that could mislead critical defect analysis [2][14][16][21][23].

B. Practical Implementation and Deployment Challenges

Despite theoretical advancements in XCT, practical industrial adoption is prevented by several limitations that are often neglected in the literature [27], such as dataset scarcity and imbalance.

Manual annotation of 3D X-ray volumes at the voxel level is labor-intensive, error-prone, and requires specialized domain expertise, creating a significant bottleneck for supervised learning paradigms [3]. AI models often exhibit domain drift, overfitting to their training data and struggling with cross-platform generalization [23]. A model trained on high-resolution data from one specific CT scanner often experiences a significant performance decrease when applied to images acquired with different exposure times, tube voltages, material densities, or focal spot drifts [6][14]. This sensitivity to device parameters necessitates continuous retraining or robust domain adaptation to ensure reliability in dynamic manufacturing environments [16].

The transition from 2D planar images to 3D volumetric XCT data exponentially increases the computational load [3]. Complex models like 3D-CNNs and Transformers require high-end GPUs and massive memory capacities, making them difficult to deploy on standard factory hardware or edge devices. The inherent conflict between millisecond-level real-time inspection and DL inference latency remains a major challenge for automated optical inspection (AOI) systems [16].

C. Future Research Directions

To bridge the gap between academic research and real-world industrial deployment, future methodologies must prioritize data-efficient and hardware-agnostic architectures [16]. Advancements in weakly supervised, semi-supervised, and unsupervised learning are essential to reduce reliance on exhaustive manual annotations [1][2][39].

The integration of PEFT and transfer learning can adapt large foundation models to specific industrial tasks using minimal annotated data [24]. Optimizing algorithms through model compression, quantization, and hardware-aware network design will be crucial for deploying robust, low-latency AI solutions directly onto edge computing platforms [16].

V. CONCLUSION

The review highlights that AI-driven interventions enhance XCT image quality and capability in microelectronics failure analysis. Despite advancements, the complete integration of cross-modal heterogeneous data remains limited by unresolved spatiotemporal synchronization challenges across multiple

sensors in dynamic environments [16], delaying migration in medical imaging and autonomous driving applications. By employing DL-based iterative reconstruction techniques, industries can reduce scan times and radiation doses while enhancing image quality and mitigating artifacts [14][42]. Ultimately, integrating AI into XCT marks a transformative change from passive structural visualization towards active diagnostic systems capable of autonomous operation, essential for the realization of the next industry generation [43].

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DISCLAIMER

The content of this material does not necessarily represent the official position of the European Union or the Government of Romania.

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